Mohammed Like

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PROFESSIONAL SUMMARY

Quantitative Analyst with hands-on experience in derivatives pricing (Black-Scholes, Monte Carlo simulation) and credit risk modeling (Basel III IRB, XGBoost, Vasicek ASRF). Built end-to-end Python pipelines for options analytics, regulatory capital estimation, and sector-level investment analysis. Targeting quant research, financial data science, and risk modeling roles where statistical rigor and financial theory intersect.

WORK EXPERIENCE

Data Analyst Intern *Visual Solutions Pvt Ltd | May 2025 – Sep 2026*

- Built a sectoral fund-flow analysis model in Python using 3-year historical CMIE/MCA dataset; surfaced 4 high-conviction capital allocation shifts, reducing manual research time by ~40%.
- Produced an investment trend report using time-series decomposition on sector-level NAV data; identified 3 cyclical regime shifts with 87% directional accuracy vs BSE sectoral indices.

AI/ML Intern *Dreamteam Technologies | May 2024 – Jun 2024*

- Engineered an NLP intent classifier (fine-tuned BERT) achieving 89% classification accuracy; reduced average analyst query-routing time by 30% across 10,000+ monthly support interactions.
- Applied sentiment analysis pipeline (VADER + logistic regression) on customer feedback corpus; improved intent accuracy by 12% over baseline rule-based system.

PROJECTS

Black-Scholes Options Pricing Engine *Python · Streamlit · SciPy · yfinance*

- Implemented Black-Scholes closed-form solver and 10,000-path Monte Carlo validator; achieved <0.3% pricing error vs theoretical benchmark across 500 NIFTY option contracts.
- Computed full Greeks portfolio (Delta, Gamma, Vega, Theta, Rho) with real-time implied volatility surface calibration using Newton-Raphson solver; validated against live NSE options data.
- Deployed interactive Streamlit dashboard with 3D vol surface visualization, scenario P&L analysis, and automated PDF report generation for risk summaries.

Credit Risk Modeling Framework (Basel III IRB) *XGBoost · Statmodels · ReportLab*

- Built Basel III IRB-compliant PD/LGD/EAD estimation pipeline on 50,000 synthetic loan records (latent variable modeling); XGBoost PD model achieved ROC-AUC 0.91, KS statistic 0.63, Gini 0.82.
- Designed WoE-based credit scorecard with PDO scaling (300–850); validated model stability via PSI monitoring and population drift analysis across 6-time windows.
- Stress-tested portfolio resilience under 3 macro scenarios (base/adverse/severe) using Vasicek ASRF model; computed regulatory capital requirements under each regime

Factor Model & Sector Rotation Analysis *Python · SQL · Power BI · Statmodels*

- Implemented Fama-French 3-factor model on NSE sectoral index data (2018–2024); decomposed alpha, beta, SMB, and HML exposures to identify momentum and value tilts across 11 sectors.
- Built STL time-series decomposition pipeline in Python to identify cyclical patterns and structural breaks in sector-level FII/DII flow data; achieved 83% accuracy in trend-change detection.
- Delivered interactive Power BI dashboard with KPI tracking and drill-down for strategic asset allocation decisions.

S K I L L S

Quantitative Methods: Options Pricing (Black-Scholes, Monte Carlo), Credit Risk (PD/LGD/EAD, Vasicek ASRF, Basel III IRB), Factor Models (Fama-French), Time-Series (ARIMA, GARCH, STL), Stochastic Modeling, Regression & Classification, Portfolio Optimization

Programming & Tools: Python (NumPy, pandas, scikit-learn, Statmodels, SciPy, XGBoost, Streamlit, yfinance), SQL, Power BI, Git, LaTeX, ReportLab

Finance & Risk: Derivatives Analytics, Risk Modeling (VaR, CVaR), Asset Pricing (CAPM, Factor Models), Backtesting, Credit Scoring, Regulatory Capital (Basel III), Implied Volatility Surfaces

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E D U C A T I O N

B. tech — Computer Science & Engineering (Data Science)

Jodhpur Institute of Engineering and Technology *2022 – 2026*

Relevant Coursework: Probability & Statistics, Financial Engineering, Machine Learning, Time Series Analysis, Optimization Techniques, Data Mining, Database Management Systems